Cenxi Zhang
Tel: +86 13561355595 | Email: zhangcenxi@mail-abc.com | GitHub: github.com/zcxixixi

EDUCATION

Qingdao University of Technology *Sep 2023 - Jun 2027*
Bachelor of Engineering in Building Electrical and Intelligent
GPA: 4.52 / 5.0 (Ranking: 3/70)

RESEARCH INTERESTS

LLM Agents; Multi-Agent Systems; AI for Embodied Interaction; Agentic Simulation; Intelligent Robotic Systems

RESEARCH, TEACHING, AND TECHNICAL EXPERIENCE

Innovation Laboratory, School of Information and Control Engineering *Jan 2024 - Present*
Student Technical Lead | Qingdao University of Technology
- Served as student technical lead for the Innovation Laboratory, coordinating AI-oriented technical training, project mentoring, and competition preparation across LLM agents, AI fundamentals, computer vision, embedded systems, and control systems.
- Mentored undergraduate students in building AI-driven perception-decision-control systems from first principles, covering model-based perception, OpenCV-based vision, PID tuning, Kalman filtering, STM32 development, and real-time debugging.
- Developed hands-on teaching workflows for AI and embedded system projects, helping students connect algorithm design, system integration, debugging, and competition-oriented engineering practice.

Friction-Aware Cooperative ADRC-NN Control for Multi-DOF Manipulators *Sep 2025 - Present*
Core Developer | MATLAB / Simulink / Python / Control Theory / Neural Networks
- Designed a cooperative ADRC and neural-network strategy for compensating nonlinear friction disturbances in multi-DOF manipulators.
- Implemented the ESO-NN control architecture and completed data collection, cleaning, feature engineering, and simulation-based validation.
- Compared the method with PID and pure ADRC baselines, focusing on tracking accuracy, robustness, and low-speed friction compensation.

32-hour AI Training Course *Apr 2026 - Jul 2026*
Developer and Presenter | Neural Networks / CNNs / Transformers / LLM Agents / Generative Models
- Developed and presented a 32-hour AI training course for 93 master's students, covering machine learning basics, neural networks, CNNs, NLP, Transformers, LLM agents, generative models, and applied AI systems.
- Created project-oriented notebooks, datasets, and code examples connecting mathematical intuition, model architecture, implementation, and deployment.
- Published course materials on GitHub and Bilibili, including lecture notebooks, code walkthroughs, and project demonstrations: github.com/zcxixixi/ai-course.

HKU Data Intelligence Laboratory nanobot Open-Source Project *Feb 2026 - Apr 2026*
Open-Source Contributor | Python / CLI / IMAP / SMTP / GitHub
- Contributed multiple pull requests to the HKU Data Intelligence Laboratory nanobot open-source project.
- Extended external communication capabilities for agents and improved CLI reliability for CJK text workflows.
- Refined interaction-stack logic for command handling and user-agent communication flow.

Embedded Vision Measurement and Control System *Aug 2025*
Team Leader, National Undergraduate Electronics Design Contest | C / Python / YOLO / OpenCV / Serial Communication | Demo: youtu.be/vVcSWguqrLc

- Led a three-member team to build a full-stack embedded vision measurement and control system from scratch within a four-day, three-night closed national contest, winning National First Prize, ranking 3rd in Shandong Province, and placing among the national top 1%.
- Developed the end-to-end vision pipeline, integrating YOLO-based object recognition, OpenCV geometric measurement, camera calibration, and perspective transformation for real-time target localization.
- Designed a side-length-variation-based differential algorithm to separate overlapping targets and improve coordinate mapping reliability.
- Implemented serial communication between the vision module and the control unit, enabling real-time data transfer for downstream control and actuation.

LLM-Based Intelligent Wheel-Legged Interactive Robot *Jun 2025*

Core Developer, Embedded Chip and System Design Competition | LVGL / OpenCV / YOLOv8 / STM32 / IoT / LLM | Demo: youtu.be/dhhkG9M1-zg

- Developed an LLM-based intelligent wheel-legged interactive robot for the Embedded Chip and System Design Competition, winning First Prize in the East China Division.
- Integrated LLM-driven interaction logic with gesture recognition, light perception, and IoT control to support multimodal human-robot interaction.
- Built STM32-based control modules for wheel-legged mobility, sensor integration, and real-time hardware interaction.
- Designed an LVGL-based interface for human-robot interaction, system status display, and interactive control.

Selected Embedded and Robotic Systems *Sep 2024 - Jun 2025*

Core Developer | STM32 / Control Systems / Computer Vision / Robotic Actuation

- Wind Pendulum Control System: developed a dual-axis control system using PID and feedforward compensation for disturbance-resistant oscillation control. Demo: youtu.be/A5QPHtyPR2g.
- High-Precision Line-Following Car: built an embedded PID control system for stable line detection, path tracking, and adaptive speed regulation under competition-oriented conditions. Demo: youtu.be/UzqoWBQiwA0.
- Two-Wheel Self-Balancing Car: implemented Kalman-filter-based attitude estimation, cascaded PID control, and motor regulation for autonomous balancing and motion control.
- Autonomous Tic-Tac-Toe Robotic Device: integrated board-state recognition, minimax-based decision logic, and robotic arm actuation to complete human-machine gameplay.

HONORS AND AWARDS

First-Class Scholarship for Outstanding Students, Qingdao University of Technology, 2024, 2025, and 2026.

National First Prize, National Undergraduate Electronics Design Contest, University Group, 2025.

First Prize, Embedded Chip and System Design Competition, East China Division, 2025.

Third Prize, Lanqiao Cup National Finals, Embedded Design and Development, 2025.

Discipline Competition Scholarship, Qingdao University of Technology, 2025.

First Prize, National Undergraduate Electronics Design Contest, Shandong Division, 2024.

SKILLS

AI Engineering: LLM agent architecture, multi-agent workflow design, tool-augmented reasoning, RAG concepts, LangGraph, prompt and workflow engineering, Claude Code, Codex.

Programming and Systems: Python, C, MATLAB, TypeScript, FastAPI, Next.js, Git/GitHub, GitHub Actions, Linux, serial communication.

Embedded and Control: STM32, embedded C, PID and cascaded control, Kalman filtering, ADRC, MATLAB/Simulink, sensor integration, real-time debugging, on-device deployment.

Computer Vision and Robotics: YOLO, OpenCV, camera calibration, perspective transformation, geometric measurement, perception-decision-control integration, robotic actuation.

Languages: Chinese (native), English (working proficiency).